EEE 304, LAB EC2

The objective of this sequence of laboratory exercises is to implement a feedback controller on a microcontroller. The problem is a simplified version of illumination control, using a light source (led) and a sensor (photo transistor).

1. Bill of Materials (BOM)

In this laboratory experiment, you will build and test a simple Digital filter on an embedded platform. The idea is to generate a few tones, filter them and observe the output by listening to the sound generated by the tones on a speaker/headphone.

In order to carry out this experiment you will be required to obtain the following:

Table 1: List of parts to be purchased

Item	Model	Quantity
Arduino	Due	1
Micro USB cable	Any android phone charger	1
Photo Transistor (sensitive to visible radiation)	Ex. 3034a Or, PT334-6C	2
Bright White LED	>3000 mcd WHITE LED	1
Resistors	10 -100 K Ohm Optional: 100-200Ohm	1 1
Set of wires	Male to Male Jumper wires	10
Breadboard	Any breadboard	1

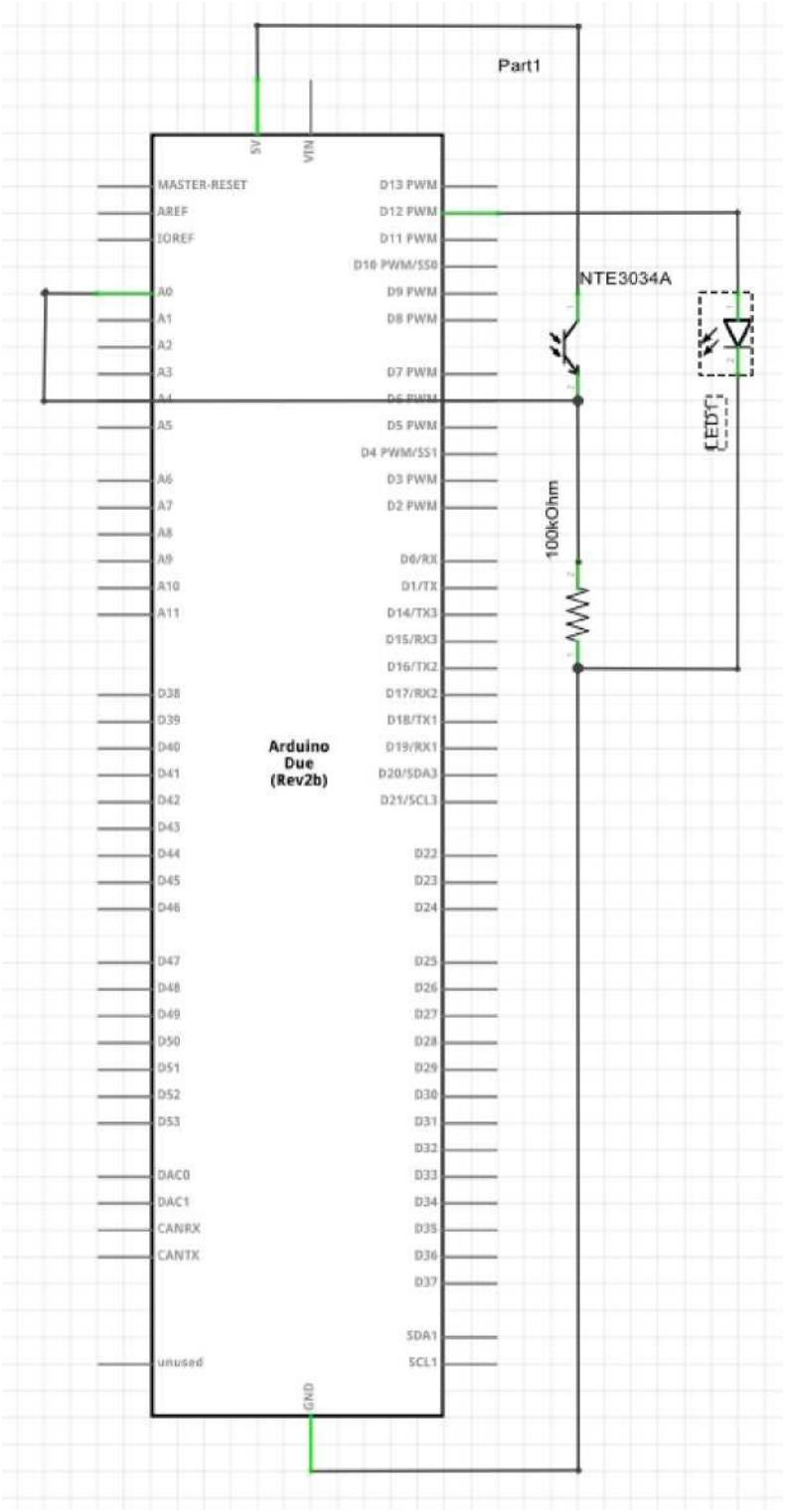
You will require at least the quantities mentioned in the table to successfully carry out the experiment. The hyperlinks attached to the **item names** are for example only and you can purchase these items from anywhere you like. In addition to the official Arduino site (www.Arduino.cc), typical vendors that carry these products are Amazon, Mouser, Digikey, Newegg, etc.

If you haven't completed the first extra credit lab assignment, you need to go through the Section 2 of Lab EC1 to make sure software is setup correctly.

2. Lab Procedure

2.1 Building the Circuit

Using the parts purchased, connect the circuit as shown in Figure 1. If you are unsure of the pin labels or for any details, you may look up datasheet for the parts online.


Part1
5V
VIN
MASTER-RESET
AREF
IOREF
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
D38
D39
D40
D41
D42
D43
D44
D45
D46
D47
D48
D49
D50
D51
D52
D53
DAC0
DAC1
CANRX
CANTX
unused
D13 PWM
D12 PWM
D11 PWM
D10 PWM/SS0
D9 PWM
D8 PWM
D7 PWM
D6 PWM
D5 PWM
D4 PWM/SS1
D3 PWM
D2 PWM
D0/RX
D1/TX
D14/TX3
D15/RX3
D16/TX2
D17/RX2
D18/TX1
D19/RX1
D20/SDA3
D21/SCL3
D22
D23
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D31
D32
D33
D34
D35
D36
D37
SDA1
SCL1
GND
Arduino
Due
(Rev2b)
NTE3034A
100kOhm
LED1

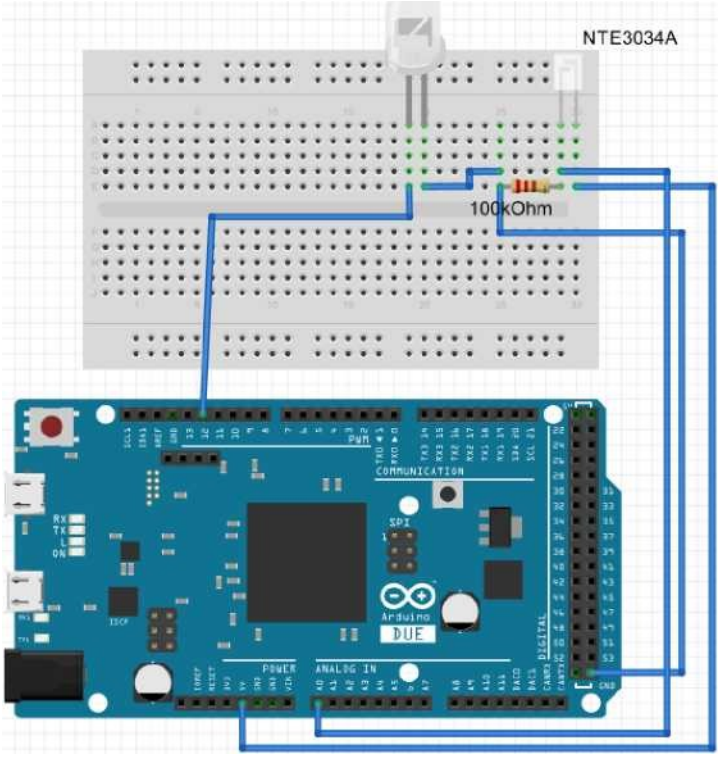


Figure 1. Building the Circuit

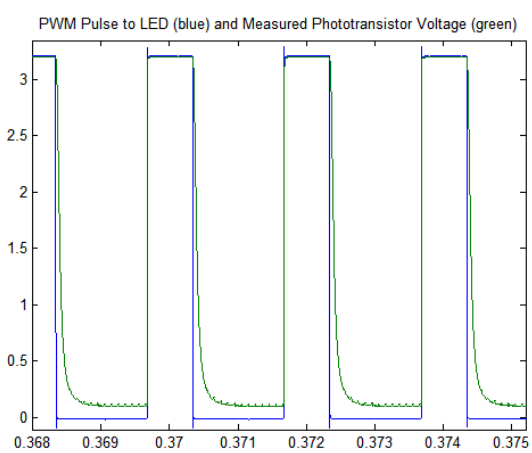
The phototransistor has two pins "emitter" (E) and "collector" (C). C goes to 5v on the Arduino and E goes to the resistor. The LED recommended emits plenty of light and is rated for 5V. So, plugging it in to the 3.3v PWM pin 12 of the Arduino will not damage the LED. A resistor 100-200Ohm may be placed between the LED and GND pin of the Arduino to limit the current. This will reduce the light output as well. You need a resistor if the LED you purchased burns out, but this is unlikely since microcontrollers do not output more than a few mA in a pin. Adding a resistor to the LED is recommended to avoid unnecessary current demands from the microcontroller, especially when many pins are active (there is a total current limit). It is also making the experiment easier in case the light intensity saturates the phototransistor too fast (for small PWM signals).

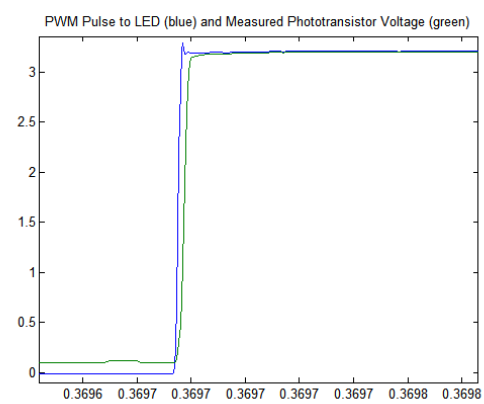
It should be kept in mind that our objective here is to create a simple experiment with minimal parts, instead of controlling a practical device. A real application would involve drivers and power amplifiers that would take the signal information from the PWM pin and convert to a signal with sufficient power to actuate the light source (Watts or kilo-Watts scale) in the specific space configuration at hand. Here, we only want to use a single sensor and a single led actuator that can be powered by the few mA that the board can generate without additional amplification. Therefore, when you place the LED and the Photoresistor on the breadboard, make sure the LED is facing the Phototransistor and that they are not too far apart. Preferably less than 1 cm distance between the two. Otherwise, the light intensity may not provide sufficient range to demonstrate the feedback control action.

2.2 Identifying the Plant

For the purposes of feedback control, we need a model for our system. From the physics of the problem, diode responds in microseconds and so does the phototransistor. Our control will therefore be limited by the sampling rate, which is in the order of kHz, so the plant model is adequately described by a gain, but it is nonlinear in terms of the applied power. An added difficulty is in the way the power is applied as a PWM input. The standard Arduino frequency of PWM is about 500Hz, meaning that every 2ms a pulse of duration 0-2ms (quantized in 8 bits, 0-255) is applied at the control pin. Depending on the characteristics of the LED, the PWM pulse may be filtered, or at least rate-limited in its increase or decrease. Then, the photo transistor has some dynamics in its response. In any case, for a constant level PWM, the measured response is not constant but changes with 500Hz frequency. At the extreme, it is similar to the PWM pulse with somewhat slower transitions.

The implication of this analysis is that the photo transistor signal must be low-pass filtered to measure the light intensity, as observed by a human. The human eye performs such an averaging and responds to frequencies in the order of 30 Hertz, an order of magnitude below the PWM frequency. Consequently, the light intensity of the LED is seen as constant. There are two ways to implement such a filtered measurement. One is to add a capacitor in parallel to the resistor so that the RC time constant is in the order of 5ms which would correspond to a filter cutoff of 32 Hz. For example, a 1uF capacitor with a 10kOhm resistor yield a 16Hz filter. That would attenuate the PWM harmonics by more than an order of magnitude so that the measured intensity looks smooth. The other approach is to filter the sampled signal, provided that the sampling rate is much faster than the PWM frequency. For example, at 10kHz the filter $F(z) = \frac{0.01}{z-0.99}$ yields a 10ms time constant. A second order filter will provide better attenuation of the PWM harmonics without reducing the cutoff frequency. Overall, in the design of the control system, one should tradeoff the variance in the measurement and the speed of the response. Then, the feedback system bandwidth (and crossover frequency) should not exceed the filter cutoff.

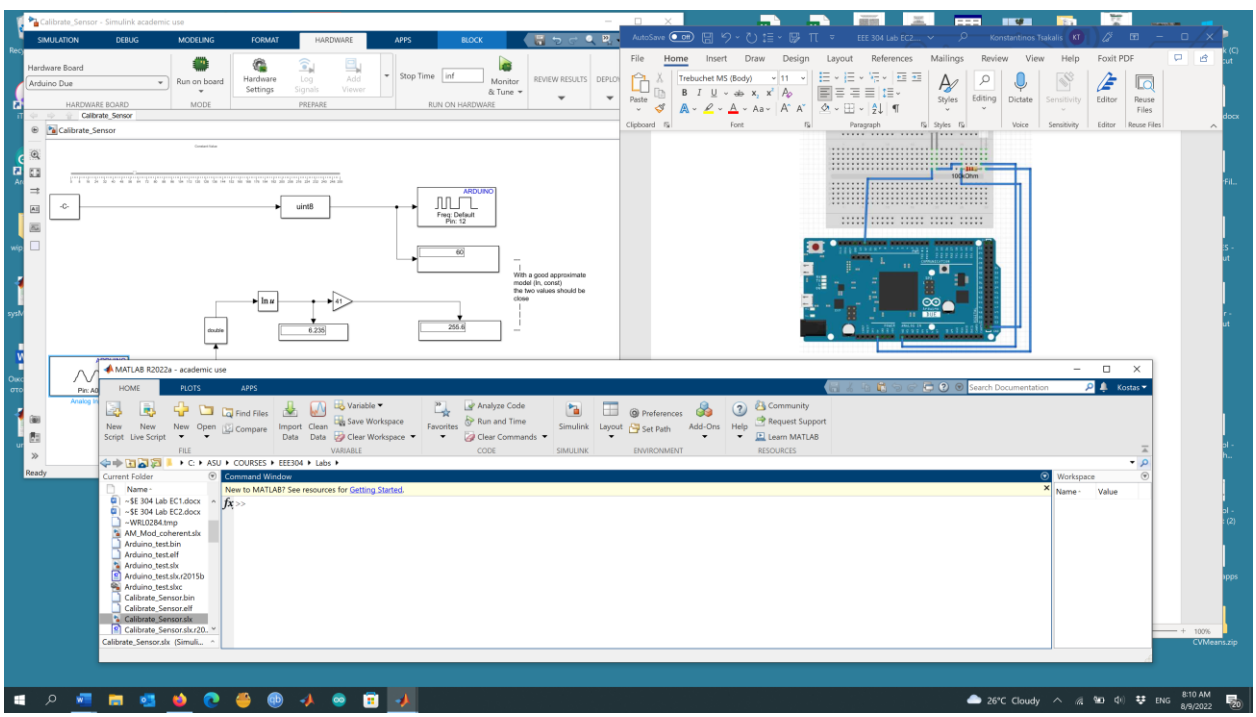




Notice that, due to the nature of the problem, the plant may be nonlinear depending on the characteristics of the LED and the phototransistor. The main objective of the feedback control is to make the plant response more predictable and linear in the external inputs. For the design of the controller, we need to perform an experiment to linearize the plant and

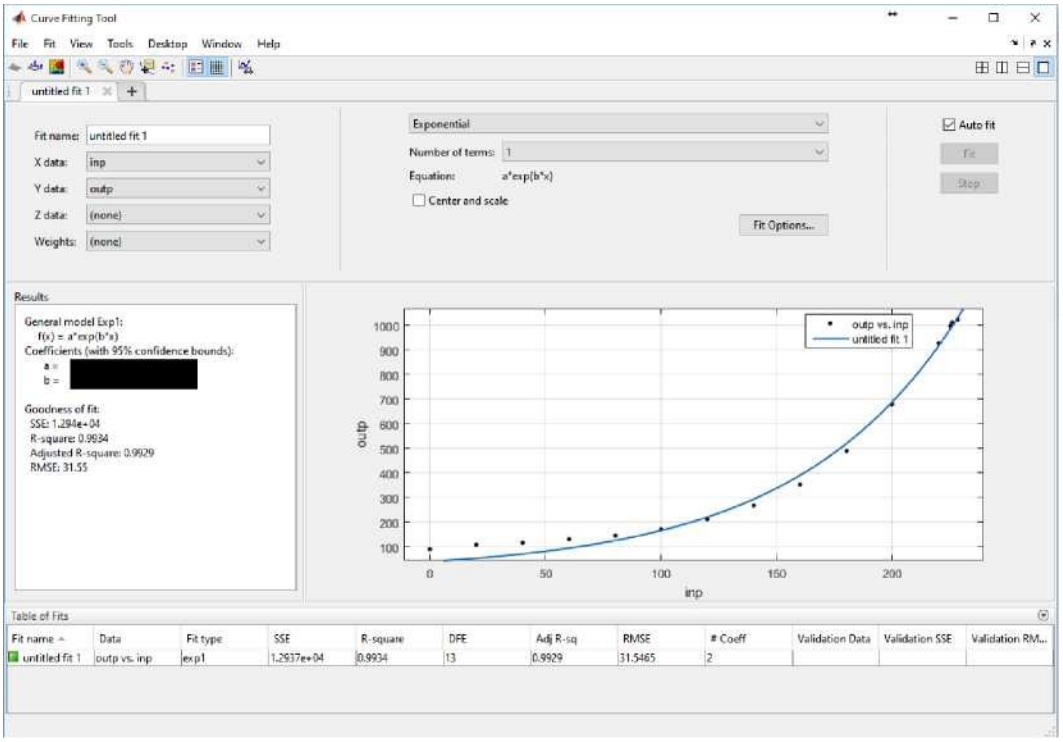
compute the effective gain of the system around the operating point. In our plant, the input is PWM voltage value which controls light intensity, and the output is light received measured as a change in current which induces a voltage on the 10k Ohm resistor. This approximation of the plant (by a linear system) is called a “linearization” and in general results in a dynamical system. But in our case a pure gain approximation is sufficient. Here the linearization is simply to find the slope of the input-output relationship with input being the PWM duty cycle and output being the light intensity observed by the ADC measurement. Note that the light intensity is measured as a change in current which induces a voltage on the 10k Ohm resistor. The relationship is nonlinear but in the vicinity of an “operating point” (e.g., 70% light intensity, or 50% PWM power) can be approximated by a straight line with slope and intercept.

Open "**Calibrate_sensor.slx**" model file. The constant value sent to the PWM pin (12), is easily adjusted by the slider. Observe, that as the values are changed between 0 - 255 in the light intensity of the LED is changing from dark/dim to bright (make sure the LED polarity is correct). Next, with the sensor correctly wired, the ADC values should also change as we change the PWM. Analog input of the Due is utilizing 12 bits (0-4095) of data that represents a measured voltage between 0-3.3V. The sampling time here is fixed 0.01sec and the more convenient execution is as Monitor&Tune, since we would like to make adjustments on the fly but we do not care to deploy this model as a standalone.



To identify a Plant model, you can use the “Curve Fitting” App from MATLAB/Simulink or any other similar tool. For the MATLAB App, create two vectors “inp” and "outp" where you record the values of the PWM and Intensities from the calibration experiment. With these variables in the workspace the data can be plotted with the "plot" command and an appropriate fitting function can be chosen. For data that seem to be mostly on a straight line except for the end points, use a linear fitting and limit the interval. For the Plant data shown below a suitable

model structure f is an exponential model $outp = a \exp(inp \times b)$. Its parameters can be estimated by linear regression of inp vs ln(outp).



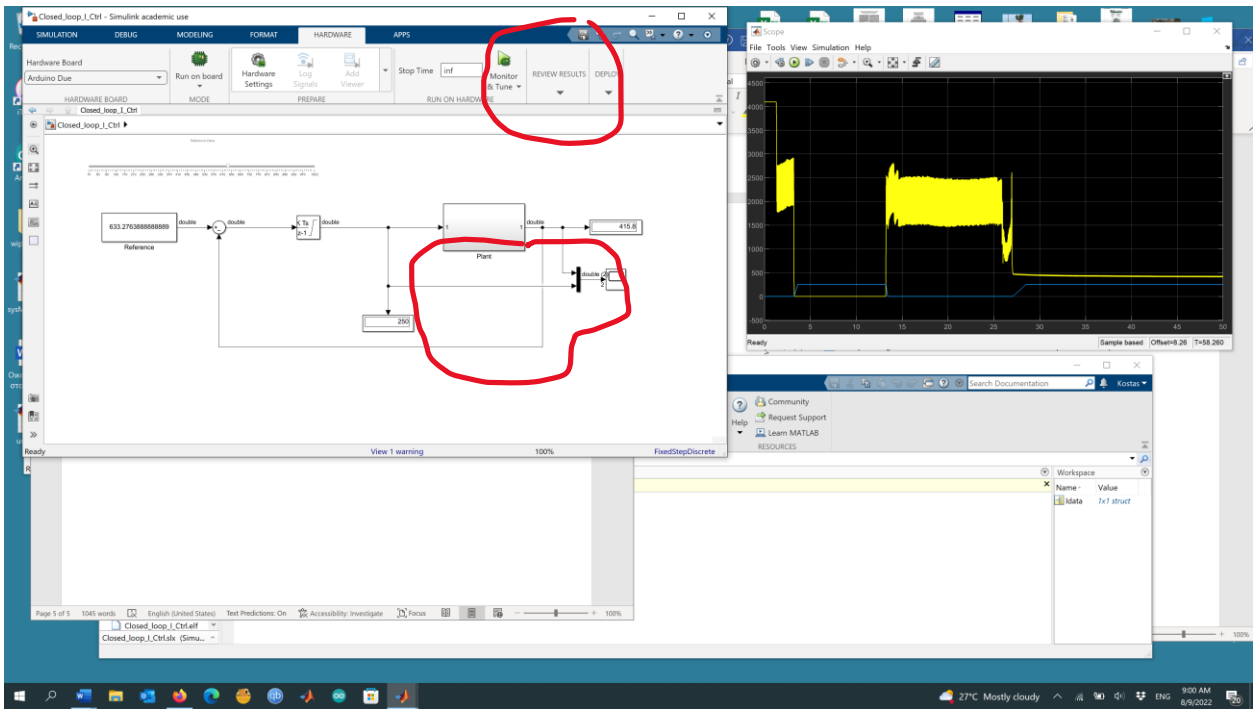
Now, we are ready to linearize the plant. Since the input-output behavior of our plant is exponential. Taking a logarithm on the output linearize it, i.e., ln*(outp)* = ln(a) + *inp* x *b*. However, we need to scale the output appropriately. This is achieved by the "gain" block. Modify the gain in the Simulink gain block to the value of "a" computed from curve fitting. Plot ln(outp) vs. inp to demonstrate that this relationship is approximately linear.

In this case we were able to determine coordinates (lin-log) where the plant is linear. In general, we may not be as fortunate. In such a case the linearization determines a linear model with bias that describes the system in the vicinity of an "operating point". A general procedure involves the Taylor series expansion of the system model: Suppose that the system i s described by y = f(x) and a particular solution y0 = f(x0) is the operating point of interest. Around that point, y-y0 = f(x)-f(x0) = f'(x0)(x-x0), where f'(x0) is the derivative of f, evaluated at x0. This is also the gain of the linearized plant that is to be used for the controller design. The offset or intercept will be handled by the integrator of the PID controller and does not play a role in the dynamics of the closed loop system. Further details on this topic are studied in subsequent courses (480, 481).

2.3 Closed-loop Control

Now that the plant is identified, we can design a controller. In our case, the plant, *P(s) = K,* a pure gain, so a simple integral controller will be sufficient to control the plant and keep the light intensity as received by the phototransistor constant. Then, the controller transfer function is (in DT) $C(z) = \frac{K_I}{z-1}$. Forming the open-loop transfer function we get $L(z) = \frac{KK_I}{z-1}$, so

the closed-loop bandwidth is determined by KK_I and the sampling time. Large values of the integral gain can cause instability, small values will cause sluggish response. Based on the theory developed in the lectures select a suitable value of the integrator gain and implement it in the controller program "**Closed_loop_I_Ctrl.slx**" (Sample Time = 0.01sec). The example shown below is representative of the screens that you see in the execution but does not correspond to an actual closed-loop control test (the sensor is not connected).



As before, using Monitor&Tune to download the program to hardware, you can perform a variety of feedback control experiments, such as changing set-points (desired illumination), adjust the speed of response, sensitivity to noise and external disturbances (other light sources).

3. Lab EC2 Assignment

#0. Write an overview of your results.

#1. Provide a table of the two vectors “inp” and "outp" you recorded as discussed in Section 2.2 with the values of the PWM and Intensities from the calibration experiment. Provide a plot and the identified values of the plant parameters (a,b).

#2. Design a controller gain and implement it in the integrator. Demonstrate how the output response is affected when this gain is changed. First, use a gain that will result in a Bandwidth of 1rad/sec (Time Constant 1sec or 100 samples for the hard-coded Sample Time of 0.01sec.) Then you can increase the gain by a factor of 10. A convenient way to demonstrate the results is to add a “small” square wave at the reference input (say 50-100 units) and take screen shots of the scope under the Monitor&Tune option.

#3. Demonstrate that if the controller gain is too large, the closed-loop system becomes unstable

#4. Take a photo of your last hardware set up and attach it to the lab report.